Advanced Multimodal Systems

Executive overview / mixed columns / raster figure

ABSTRACT

This five-page synthetic dossier intentionally combines difficult PDF structures: nested headers, rotated labels, embedded raster imagery, vector plots, and multi-column narratives. All measurements are illustrative rather than empirical.

1. Research objectives

The benchmark assesses document understanding across heterogeneous visual and textual modalities.

Primary challenges include:

- Reading-order reconstruction
- Table-cell association
- Figure-caption grounding
- Metadata preservation

2. Pipeline assumptions

Input documents may include scans, vector artwork, nested tables, and nonuniform reading directions.

Expected outputs:

- Structured JSON records
- Linked figure descriptions
- Chunk-level provenance
- Layout-aware Markdown

FIGURE 1. Embedded raster microscopy-style illustration

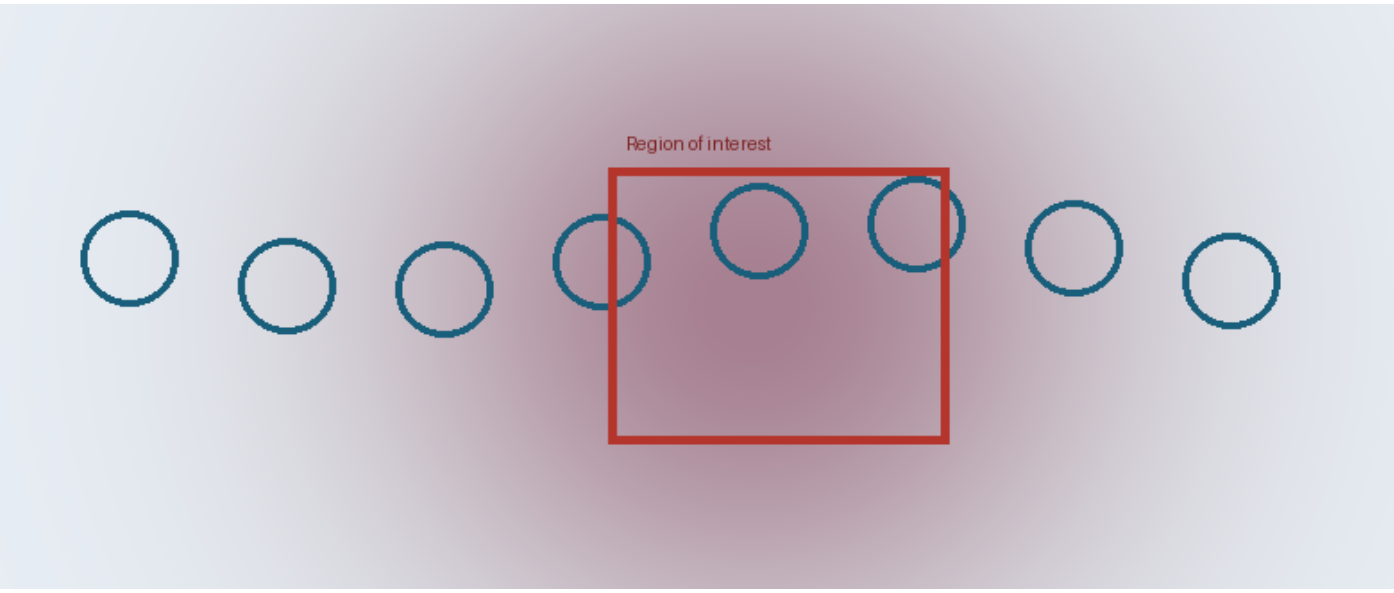


Figure 1. Synthetic pixel image with an annotated region of interest (ROI).

Document properties: five distinct page layouts; searchable vector text; one true raster image; vector diagrams and quantitative visualizations.

Experimental Matrix

Multi-level headers / merged regions / dense numerical table

TABLE 1. Hierarchical performance matrix

Method	Quality			Latency		Cost
Parser	Text F1	Table F1	Image F1	P50 ms	P95 ms	USD/1k
Rule-only	0.91	0.62	0.34	82	215	0.08
OCR hybrid	0.94	0.79	0.61	134	388	0.41
Layout model	0.96	0.88	0.83	296	720	0.77
VLM assisted	0.97	0.92	0.95	890	1810	2.65
Ensemble	0.98	0.95	0.96	1020	2300	3.12
Fallback	0.88	0.57	0.29	64	190	0.05

TABLE 2. Irregular cells with wrapped conceptual content

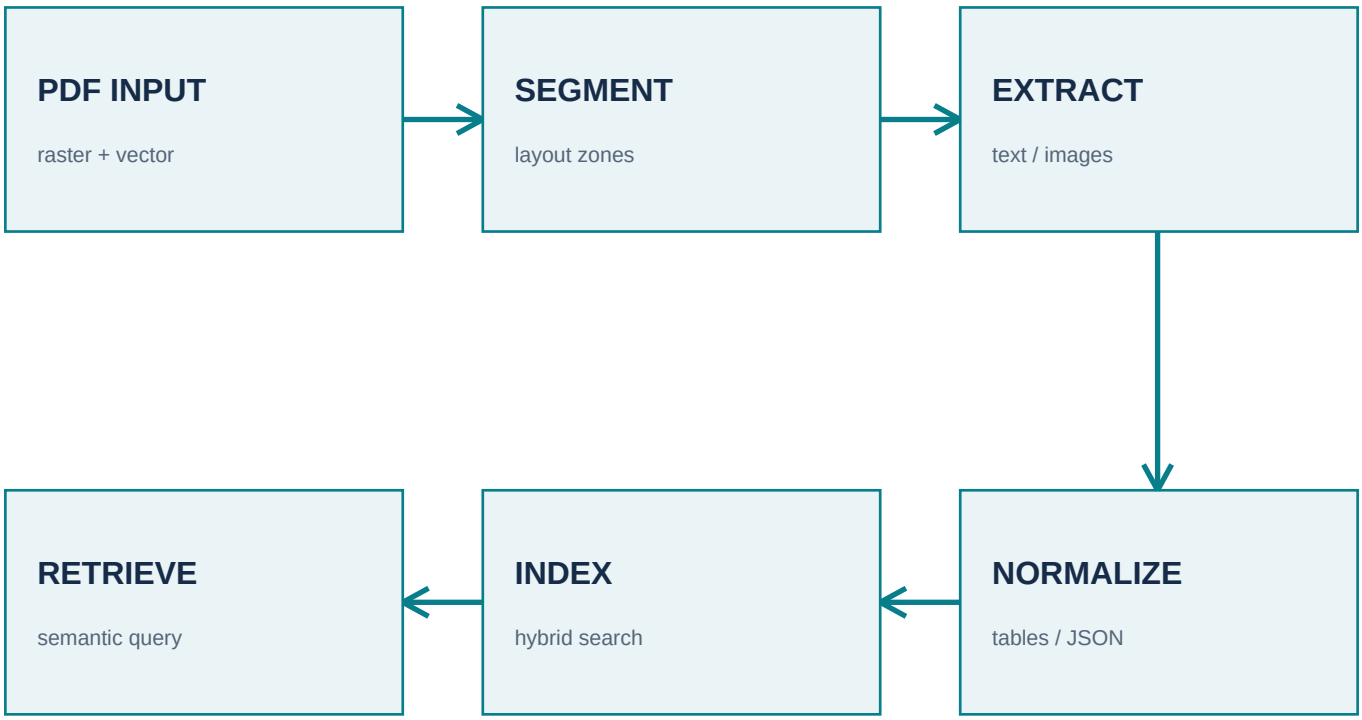
Condition / group	Observed artifact	Mitigation
A: Multi-column	Cross-column text jumps	Geometry sort
B: Spanning cells	Header-child ambiguity	Cell graph
C: Rotated labels	Orientation mismatch	Angle detect
D: Embedded figure	Caption detached	Region linking

Note: All benchmark values are invented for parser evaluation.

Architecture & Connectivity

Vector process diagram / connectors / annotations / legends

FIGURE 2. Multi-stage document processing graph



Vector symbols and parsing requirements

01 Directed arrows encode dependency

02 Each node contains hierarchical text

03 Spatial adjacency differs from reading order

04 Boxes and connectors are vector graphics

Expected structural relation:

INPUT -> SEGMENT -> EXTRACT -> NORMALIZE -> INDEX -> RETRIEVE

Quantitative Figures

Line plot / grouped bars / axis labels / embedded legend

FIGURE 3. Quality and throughput across document complexity

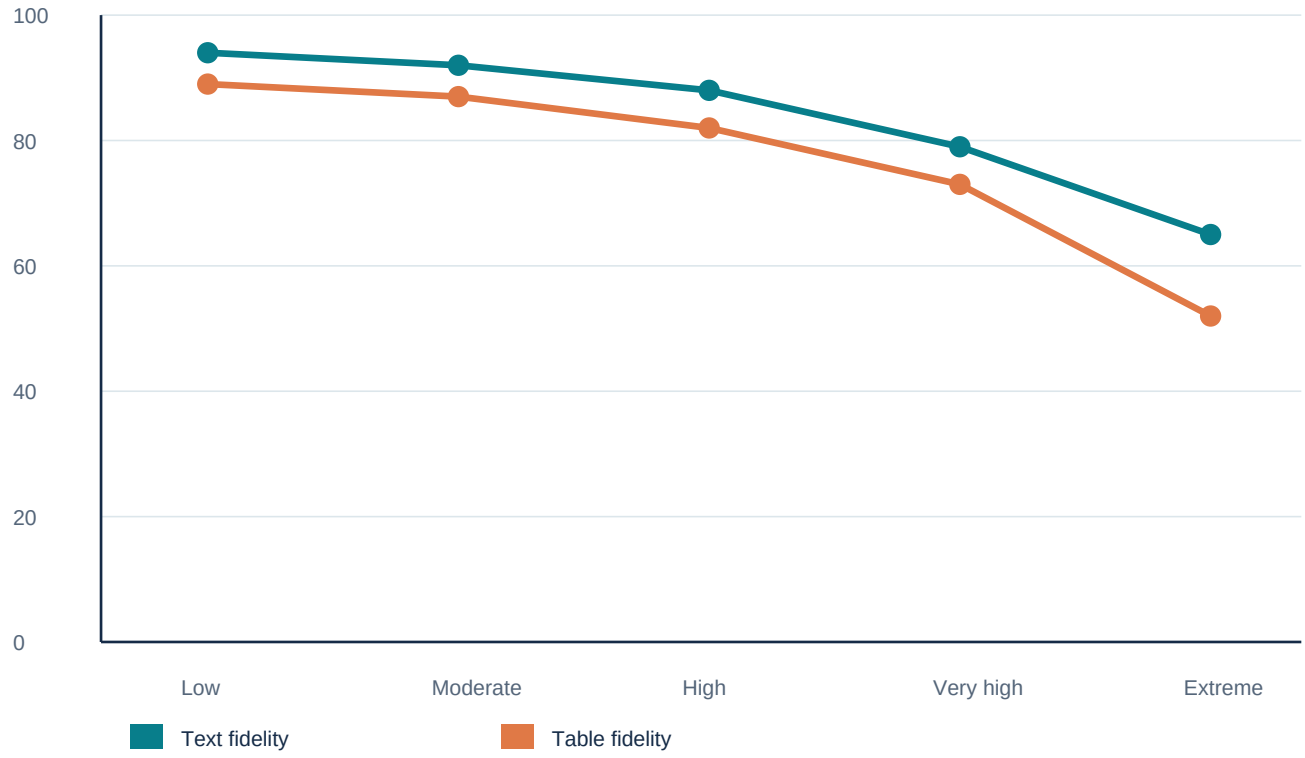


FIGURE 4. Processing time by pipeline stage

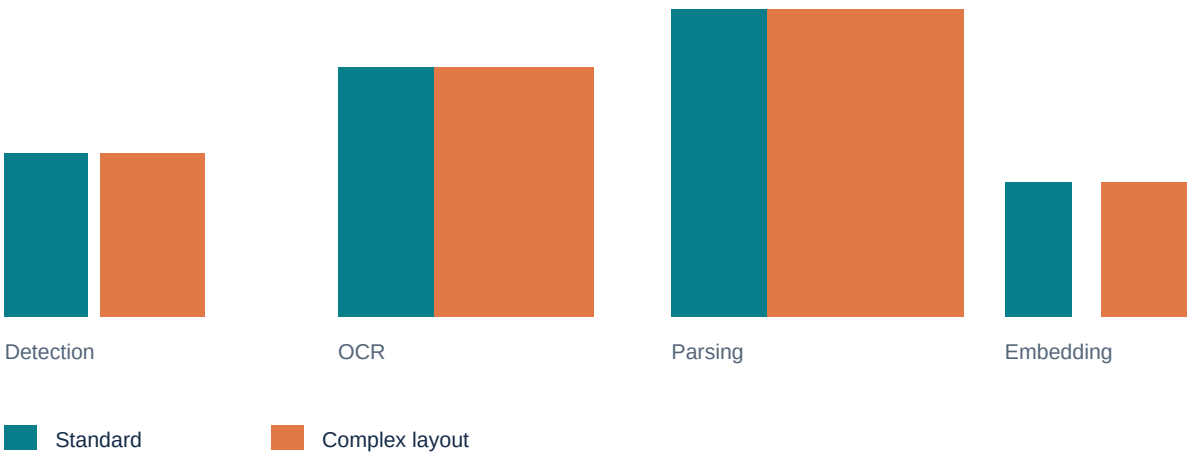


Figure 3-4. Synthetic values. Vector-drawn plots are intentionally not bitmap charts.

Appendix: Edge Cases

Rotated text / two-column reading order / checklist / mixed typography

A. Extraction validation checklist

- [ ] Detect merged header cells and infer parent-child relationships
- [ ] Preserve table column order, units, and numeric precision
- [ ] Associate captions with nearby figures and raster assets
- [ ] Recover multi-column paragraphs without interleaving lines
- [ ] Capture arrows and nodes as a structured directed graph
- [ ] Retain page number, bounding box, and source provenance

B. Rotated annotation and dense layout

ROTATED SIDE LABEL / 90 DEGREES

Column A - embedded narrative
A layout-aware parser should
retain this contiguous block
despite nearby orthogonal
text and graphics.

Column B - metadata
Document ID: DEMO-005
Revision: 2.1 (synthetic)
Confidence: 0.973
Reading order: A then B

C. Structured output example

```
{
  "page": 5, "type": "mixed_layout",
  "regions": ["checklist", "rotated_label", "columns"],
  "table_count": 0, "figure_count": 0,
  "requires_geometry": true
}
```